

BACKUP AND RESTORE(TAR & GZIP)

TASKS

1.Create a backup of your Music folder (or any folder with at least 3 files) using the tar command and compress it with gzip, saving the output as music_backup.tar.gz in your home directory.

Create a Backup of Music Folder

Step 1: Go to Home Directory

```
cd /home/du
```

Step 2: Create Music Folder

```
mkdir Music
```

Step 3: Create 3 Sample Files

```
touch Music/song1.mp3  
touch Music/song2.mp3  
touch Music/song3.mp3
```

Verify:

```
ls Music
```

```
[root@localhost du]# ls Music  
song1.mp3 song2.mp3 song3.mp3  
[root@localhost du]#
```

Step 4: Create Backup

```
tar -czvf music_backup.tar.gz Music
```

```
[root@localhost du]# tar -czvf music_backup.tar.gz Music  
Music/  
Music/song1.mp3  
Music/song2.mp3  
Music/song3.mp3  
[root@localhost du]#
```

Step 5: Verify Backup

ls -lh

```
[root@localhost du]# ls -lh
total 4.0K
drwxr-xr-x. 2 du du 6 Jul 13 14:10 Desktop
drwxr-xr-x. 2 du du 6 Jul 13 14:10 Documents
drwxr-xr-x. 2 du du 26 Jul 14 11:52 Downloads
drwxr-xr-x. 2 du du 57 Jul 15 21:02 Music
-rw-r--r--. 1 root root 185 Jul 15 21:03 music_backup.tar.gz
drwx-----. 2 du du 24 Jul 14 13:39 mysite
drwxr-xr-x. 2 du du 6 Jul 13 14:10 Pictures
drwxr-xr-x. 2 du du 6 Jul 13 14:10 Public
drwxr-xr-x. 2 du du 6 Jul 13 14:10 Templates
drwxr-xr-x. 2 du du 6 Jul 13 14:10 Videos
[root@localhost du]#
```

2.Extract the contents of a tar.gz file (for example, music_backup.tar.gz) into a new directory called restored_music and verify that all files are restored correctly.

Restore Backup

Step 1: Create Restore Folder

mkdir restored_music

Step 2: Extract Backup

tar -xzvf music_backup.tar.gz -C restored_music

```
[root@localhost du]# tar -xzvf music_backup.tar.gz -C restored_music
Music/
Music/song1.mp3
Music/song2.mp3
Music/song3.mp3
[root@localhost du]#
```

Step 3: Verify

ls restored_music/Music

```
[root@localhost du]# ls restored_music/Music
song1.mp3 song2.mp3 song3.mp3
[root@localhost du]#
```

3. Write a shell command that creates a compressed backup of all .jpg images from your Downloads folder into a file named images_backup.tar.gz, excluding any files larger than 5MB.
Hint: Use the find command with size filtering and tar with -files-from.

Backup Only JPG Files

Step 1: Create Downloads Folder

```
mkdir Downloads
```

Step 2: Create Sample Files

```
touch Downloads/image1.jpg  
touch Downloads/image2.jpg  
touch Downloads/image3.jpg  
touch Downloads/file.txt
```

Verify

```
ls Downloads
```

Step 3: Create Backup of JPG Files Only

```
find Downloads -name "*.jpg" -size -5M | tar -czvf images_backup.tar.gz -T -
```

```
[root@localhost du]# touch Downloads/file.txt  
[root@localhost du]# ls Downloads  
file.txt  flipkart.jpg  image1.jpg  image2.jpg  image3.jpg  
[root@localhost du]# find Downloads -name "*.jpg" -size -5M | tar -czvf images_b  
ackup.tar.gz -T -  
Downloads/flipkart.jpg  
Downloads/image1.jpg  
Downloads/image2.jpg  
Downloads/image3.jpg  
[root@localhost du]#
```

Step
4:
Verify

Backup

```
tar -tvf images_backup.tar.gz
```

```
Downloads/image3.jpg
[root@localhost du]# tar -tvf images_backup.tar.gz
-rw-r--r-- du/du          7688 2026-07-14 11:52 Downloads/flipkart.jpg
-rw-r--r-- root/root      0 2026-07-15 21:13 Downloads/image1.jpg
-rw-r--r-- root/root      0 2026-07-15 21:13 Downloads/image2.jpg
-rw-r--r-- root/root      0 2026-07-15 21:14 Downloads/image3.jpg
[root@localhost du]#
```

4. Simulate a restore scenario: delete one file from your original Music folder, then use your music_backup.tar.gz archive to restore only the deleted file without overwriting the rest.

Restore Deleted File

Step 1: Delete One File

```
rm Music/song2.mp3
```

Verify:

```
ls Music
```

```
[root@localhost du]# ls Music
song1.mp3  song3.mp3
[root@localhost du]#
```

Step 2: Restore Only Deleted File

```
tar -xzvf music_backup.tar.gz Music/song2.mp3
```

```
song1.mp3  song3.mp3
[root@localhost du]# tar -xzvf music_backup.tar.gz Music/song2.mp3
Music/song2.mp3
[root@localhost du]#
```

Step

3: Verify

```
ls Music
```

```
[root@localhost du]# ls Music
song1.mp3  song2.mp3  song3.mp3
[root@localhost du]#
```